

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2023-24

Paper No. MTB-401: Partial Differential Equations

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074879

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

- (a) Form the PDE by eliminating h and k from $(x - h)^2 + (y - k)^2 + z^2 = r^2$. [2]
- (b) Form the PDE by eliminating arbitrary functions f and g from $z = f(x + at) + g(x - at)$. [2]
- (c) Define the order and degree of a PDE. Give an example. [2]
- (d) Find the complete integral of $z = px + qy + p^2 + q^2$, where $p = \partial z / \partial x$, $q = \partial z / \partial y$. [2]
- (e) Solve $(D_x^2 - 4D_x D_y + 4D_y^2)z = 0$, where $D_x = \partial / \partial x$, $D_y = \partial / \partial y$. [3]
- (f) Find the general solution of $s + p - q = z$. [3]
- (g) Solve $x^2 r - y^2 t + xp - yq = 0$ by reducing to linear PDE with constant coefficients. [3]
- (h) Solve $xr + p = 9x^2 y^3$. [3]
- (i) Check whether $3r + 6s + 4t = 0$ is elliptic. [2]

Question 2.

- (a) Obtain the general integral of $pt + 3q = 5z + \tan(y - 3x)$. [5]
- (b) Apply Charpit's method to find the complete integral of $px + qy + z = xq^2$. [7]

Question 3.

- (a) Find the general integral of $r + s - 6t = y \cos x$. [5]
- (b) Reduce $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial y} = x^2 y^2$ to linear PDE with constant coefficients and solve it. [7]

Question 4.

(a) Solve $(2D_x^2 - 5D_xD_y + 2D_y^2)z = 24(y - x)$. [5]

(b) Solve $(D_x^2 - D_xD_y - 2D_y^2 + 2D_x + 2D_y)z = e^{x+y}$. [7]

Question 5.

(a) Solve $ys - p = xy^2 \cos(xy)$. [5]

(b) Reduce to canonical form and solve: $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$. [7]

Question 6.

(a) Solve by Monge's method: $yr - 2ys + t = p + q^2$. [5]

(b) Solve $r + 4s + t + rt - s^2 = 2$. [7]

Question 7.

(a) Use separation of variables to solve the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$, subject to $u(x, 0) = x^2 - x$, $u(0, t) = u(1, t) = 0$. [6]

(b) Solve the wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ using separation of variables. [6]